

TrES-3b

R-2282

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120515 · created 2026-08-17T00:13:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456062.90253 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.019
Transit depth (Rp/Rs)^2	3.86 +/- 0.74 [%]
Orbital Inclination (inc)	81.93 +/- 0.5
Airmass coefficient 1 (a1)	1.241 +/- 0.021
Airmass coefficient 2 (a2)	-0.21 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.7 %
Best Comparison Star	None
Optimal Aperture	3.88
Optimal Annulus	13.72
Transit Duration (day)	0.0593 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.019 vs 0.16309 (deviation 1.01σ)
Duration fit vs published	0.0593 d vs 0.0567 d (deviation 0.32σ)
Mid-time O–C	1.8 min
Depth SNR	5.9
Residual scatter	1.7 %

Light curve

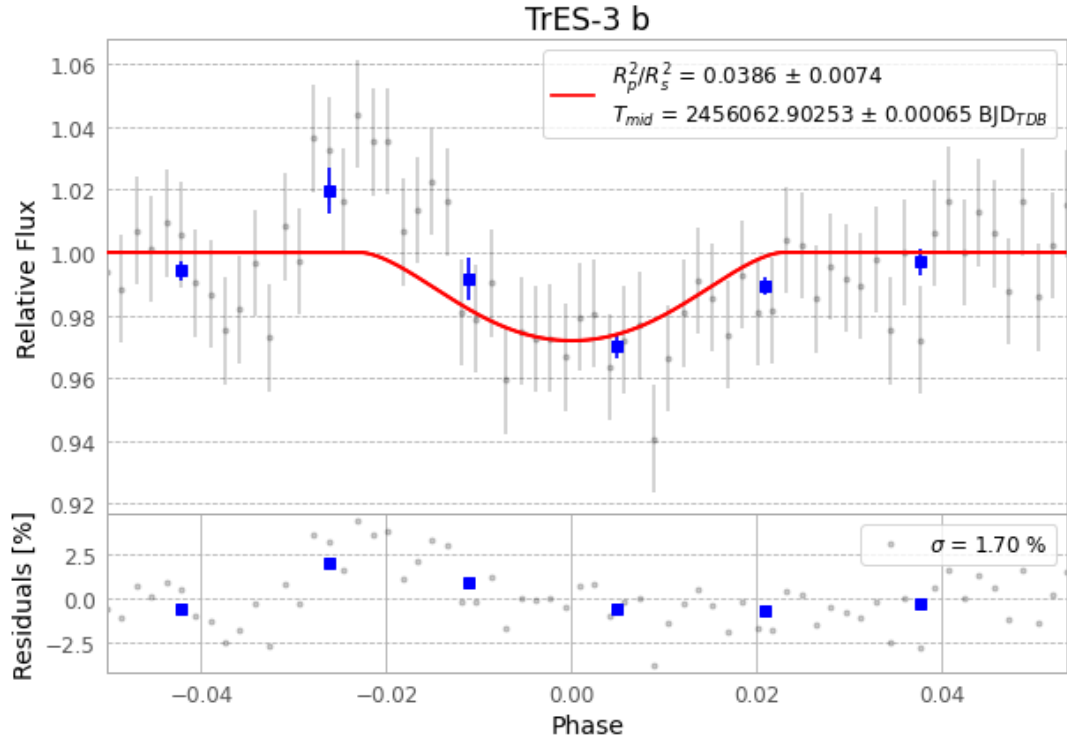


Figure 1 — FinalLightCurve_TrES-3 b_2012-05-15.png (SHA-256 5548d898552b1386...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 260], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8b35bf7faed58174...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

66 FITS frames (43 MB), TRES-3120515080013.FITS → TRES-3120515111513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120515.log (33dce565b57b...), FinalLightCurve_TrES-3 b_2012-05-15.png (5548d898552b...), FinalLightCurve_TrES-3 b_2012-05-15.csv (e100b6469ffe...)

Compute resources

Wall time 210.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:13Z.